

Incident Report: Unauthorized Cloud Infrastructure Access & Endpoint Anomalies

Project: BOTSv3 Incident Analysis (COMP3010)

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Reporting Authority: Chief Information Security Officer (CISO), Frothly Security Operations Center

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<https://github.com/Mattfish/COMP3010-SecOps-and-Incident-Management.git>

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1. Introduction

This report presents a forensic investigation into security anomalies within Frothly's cloud and endpoint infrastructure using the BOTSv3 dataset in Splunk. The scope is limited to AWS CloudTrail, S3 access logs, and Windows endpoint telemetry, with a focus on identifying misconfigurations and behaviours consistent with insider misuse or credential compromise.

The primary audience is the CISO and SOC team, and the report is written as an internal incident record to inform SOC process improvements.

The report is structured as follows: Section 2 links the investigation to SOC roles and the NIST incident lifecycle; Section 3 documents Splunk setup and dataset ingestion; Section 4 provides guided analysis answering the BOTSv3 question set; Section 5 consolidates conclusions and presents a prioritised action plan.

2. SOC Roles & Incident Handling Methodology

Security Operations Centers (SOC) rely on a tiered structure to manage alert fatigue and ensure critical threats are escalated appropriately [1].

- Tier 1 Analysts (Triage): Monitor SIEM alerts to distinguish false positives from genuine security events.
- Tier 2 Analysts (Incident Responders): This investigation assumes the role of a Tier 2 analyst. The focus is not merely viewing alerts, but correlating data across multiple sourcetypes (AWS, Endpoint, Network) to reconstruct the attack timeline.
- Tier 3 Analysts (Threat Hunters): Proactively search for threats that evade automated detection rules.

2.1 Application of the NIST Cycle

BOTSv3 simulates Frothly's production environment, including AWS web workloads, S3 storage, and corporate endpoints. Aligned with the NIST SP 800-61 Rev 3 lifecycle, this investigation aims to [2]:

- Identify & Protect: Audit user accounts and endpoints to uncover 'Least Privilege' violations.
- Detect & Respond: Analyze logs to determine exactly how an S3 bucket was exposed, who was responsible, and what data was at risk.
- Recover & Govern: Assess the failures in configuration and governance that caused the breach to propose strategic improvements for future resilience.

3. Environment Setup & Dataset Ingestion

3.1 Environment Setup

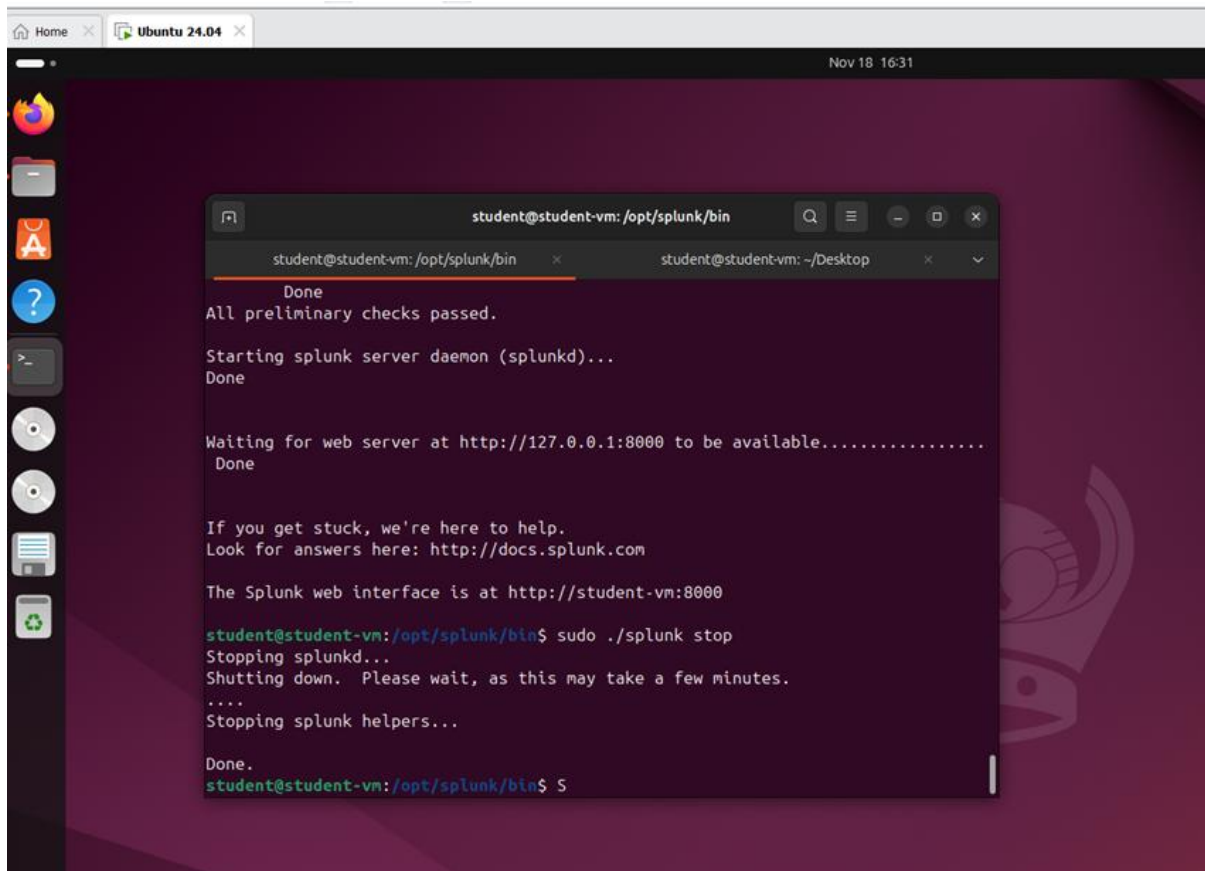
A local Splunk Enterprise instance was deployed on an Ubuntu virtual machine to mimic how a SOC would typically host a centralised SIEM in a controlled environment. The VM approach isolates the BOTSv3 lab from production resources while still allowing realistic log volume and query performance. Splunk was installed under /opt to follow Linux best practice for third-party software and keep application files separate from the OS.

Commands used for installation

```
wget -O splunk-8.x.x-linux-2.6-amd64.tgz 'https://download.splunk.com/...'
```

```
sudo tar -xvzf splunk-*.tgz -C /opt
```

```
sudo /opt/splunk/bin/splunk start --accept-license
```

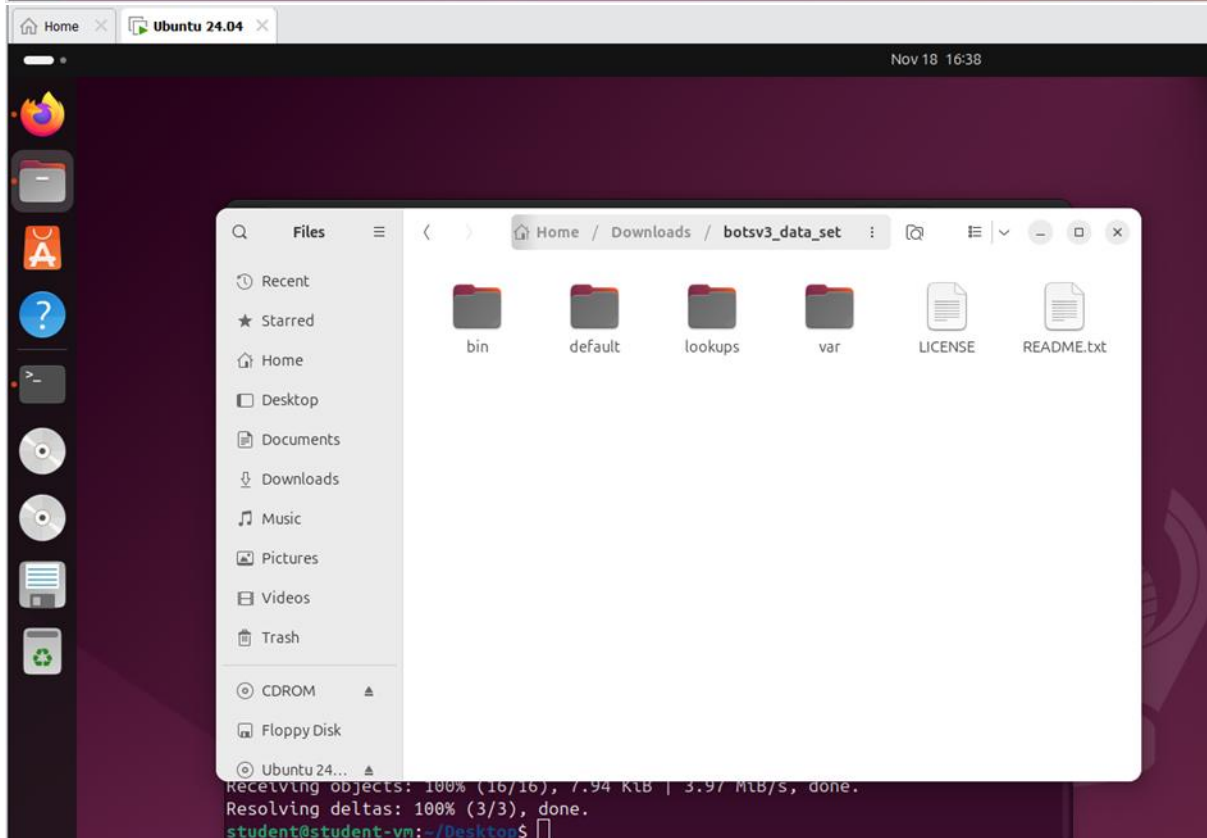
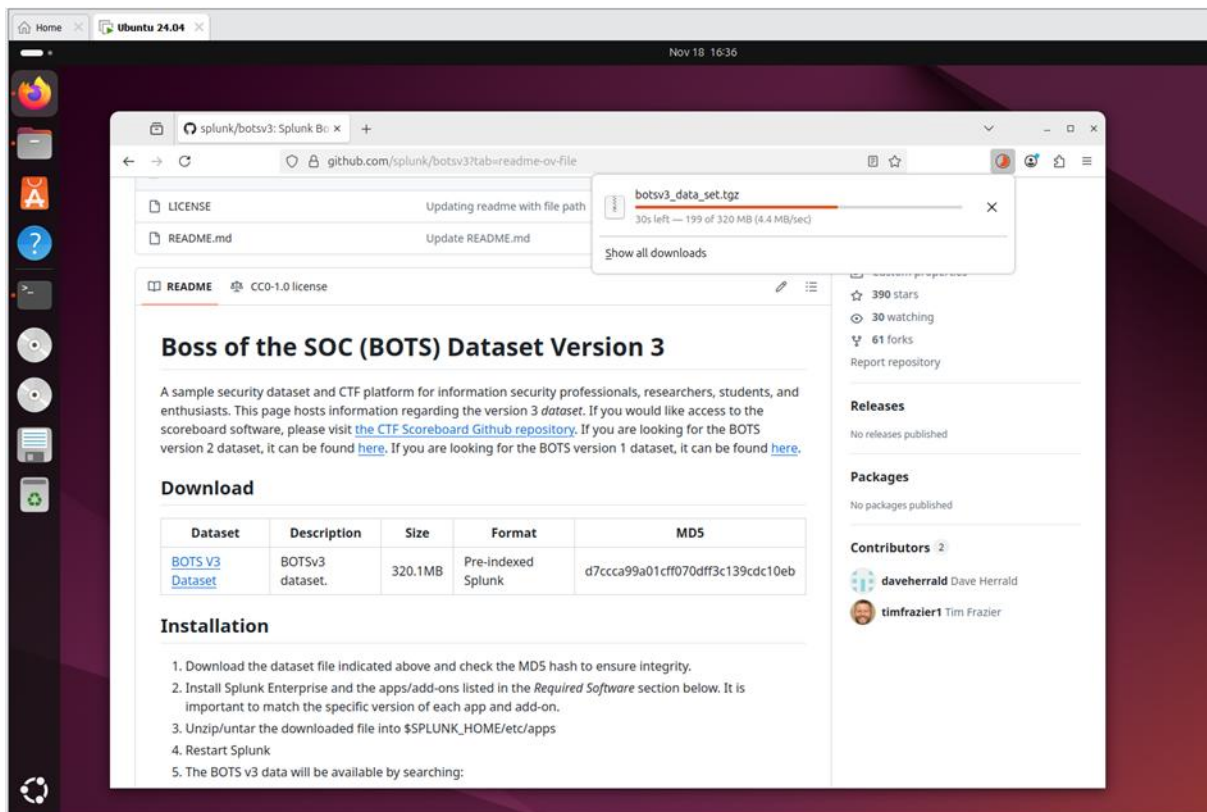


The screenshot shows a terminal window titled 'student@student-vm: /opt/splunk/bin' running on an Ubuntu 24.04 desktop. The terminal output shows the successful installation and startup of Splunk. The user has run 'sudo ./splunk start --accept-license' and the output indicates that all preliminary checks passed and the Splunk server daemon (splunkd) is running. The terminal also shows the user running 'sudo ./splunk stop' to stop the service. The desktop background is the Ubuntu logo, and the terminal window has a dark theme.

```
student@student-vm: /opt/splunk/bin
Done
All preliminary checks passed.
Starting splunk server daemon (splunkd)...
Done
Waiting for web server at http://127.0.0.1:8000 to be available.....
Done
If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com
The Splunk web interface is at http://student-vm:8000
student@student-vm:/opt/splunk/bin$ sudo ./splunk stop
Stopping splunkd...
Shutting down. Please wait, as this may take a few minutes.
....
Stopping splunk helpers...
Done.
student@student-vm:/opt/splunk/bin$ S
```

3.2 Dataset Ingestion

The BOTSv3 dataset was downloaded from the official Splunk GitHub repository and installed as a dedicated app, with all events indexed into a separate botsv3 index. This separation prevents test data from polluting the default index and allows focused searches using `index=botsv3` across multiple sourcetypes such as `aws:cloudtrail`, `aws:s3:accesslogs`, `WinHostMon`, `hardware` and `access_combined`. Basic validation was performed by checking index event counts, confirming that expected sourcetypes were present, and running sample searches to ensure CloudTrail, S3 and endpoint logs were ingested correctly before starting the investigation.



4.1 IAM Account Auditing (Question 1)

Objective: Identify all IAM users accessing AWS services to audit for Least Privilege violations.

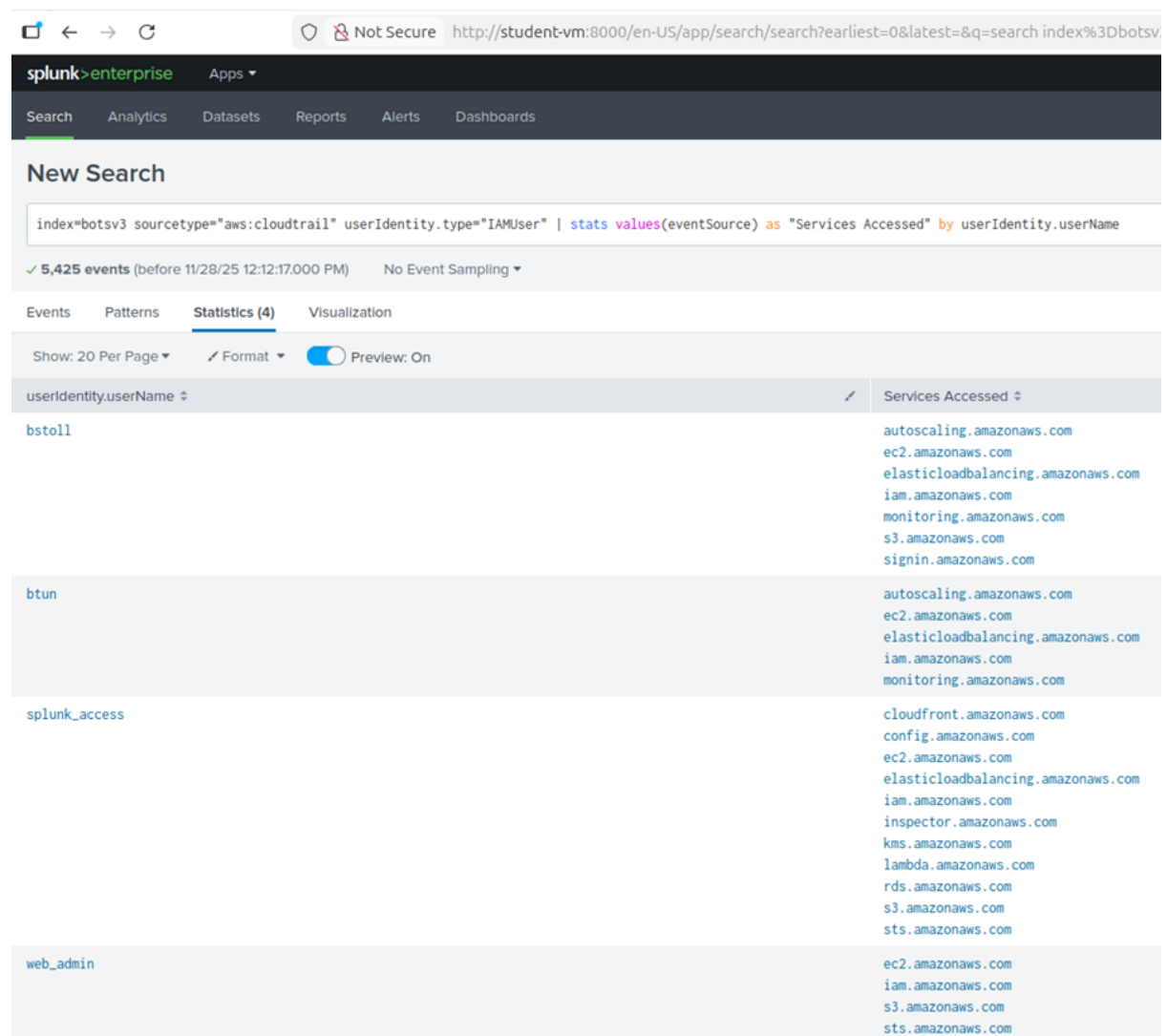
Analysis: Querying `aws:cloudtrail` for `userIdentity.type="IAMUser"` allows us to isolate human actors from automated service roles. The `stats` command grouped these by username.

Query:

```
index=botsv3 sourcetype="aws:cloudtrail" userIdentity.type="IAMUser"
```

```
| stats values(eventSource) as "Services Accessed" by userIdentity.userName
```

Finding: The users `bstoll`, `btun`, `splunk_access`, and `web_admin` were identified accessing AWS services. Knowing exactly which IAM users access AWS services allows the SOC to baseline normal behaviour, detect misuse of generic accounts like `web_admin`, and investigate suspicious or dormant accounts that suddenly become active.



The screenshot shows the Splunk Enterprise search interface. The search bar contains the query: `index=botsv3 sourcetype="aws:cloudtrail" userIdentity.type="IAMUser" | stats values(eventSource) as "Services Accessed" by userIdentity.userName`. The results show 5,425 events. The table displays the username and the services accessed.

userIdentity.userName	Services Accessed
bstoll	autoscaling.amazonaws.com ec2.amazonaws.com elasticloadbalancing.amazonaws.com iam.amazonaws.com monitoring.amazonaws.com s3.amazonaws.com signin.amazonaws.com
btun	autoscaling.amazonaws.com ec2.amazonaws.com elasticloadbalancing.amazonaws.com iam.amazonaws.com monitoring.amazonaws.com
splunk_access	cloudfront.amazonaws.com config.amazonaws.com ec2.amazonaws.com elasticloadbalancing.amazonaws.com iam.amazonaws.com inspector.amazonaws.com kms.amazonaws.com lambda.amazonaws.com rds.amazonaws.com s3.amazonaws.com sts.amazonaws.com
web_admin	ec2.amazonaws.com iam.amazonaws.com s3.amazonaws.com sts.amazonaws.com

4.2 MFA Compliance Check (Question 2)

Objective: Detect AWS API activity performed without Multi-Factor Authentication (MFA).

Analysis: To identify AWS API activity occurring without MFA, I performed a keyword search using `mfa` against the `aws:cloudtrail` sourcetype. I explicitly excluded `ConsoleLogin` events to isolate programmatic API calls from web interface logins. This revealed the nested JSON path `userIdentity.sessionContext.attributes.mfaAuthenticated`.

Finding: 2,155 events were generated with `mfaAuthenticated=false`. This volume of non-MFA activity represents a violation of the Protect function of the NIST CSF (Identity Management). In a Zero Trust architecture, the assumption that a valid API key implies a trusted user is known as "Implicit Trust"—a vulnerability that allowed the attacker to bypass verification. Monitoring this field enables the SOC to build alerts for high-risk API activity, prioritise incident response on compromised keys, and drive enforcement of stronger authentication policies [6].

Query:

```
index=botsv3 sourcetype="aws:cloudtrail" eventName!="ConsoleLogin" | stats count by  
userIdentity.sessionContext.attributes.mfaAuthenticated
```

index=botsv3 sourcetype="aws:cloudtrail" eventName!="ConsoleLogin" *mfa*

✓ 2,338 events (before 11/28/25 12:52:22.000 PM) No Event Sampling ▼

Events (2,338) Patterns Statistics Visualization

Timeline format ▼ Zoom Out Zoom to Selection Deselect

Format ▼ Show: 20 Per Page ▼ View: List ▼

Time	Event
8/20/18 4:15-20.000 PM	{ [-] awsRegion: us-west-1 eventID: 97c6bfc3-c3cf-437c-8b05-4043635ce306 eventName: DescribeInstanceStatus eventSource: ec2.amazonaws.com eventTime: 2018-08-20T15:15:20Z eventType: AwsApiCall eventVersion: 1.05 recipientAccountId: 622676721278 requestID: f4bd4e9b-e27c-4a52-93fa-fab7a76d3639 requestParameters: { [+] } responseElements: null sourceIPAddress: autoscaling.amazonaws.com userAgent: autoscaling.amazonaws.com userIdentity: { [-] accountId: 622676721278 arn: arn:aws:sts::622676721278:assumed-role/AWSSE invokedBy: autoscaling.amazonaws.com principalId: AROAIOHK7E4SHKYSVVYLM:AutoScaling sessionContext: { [-] attributes: { [-] creationDate: 2018-08-20T15:09:21Z mfaAuthenticated: false } } }

Splunk Enterprise

New Search

index=botsv3 sourcetype="aws:cloudtrail" eventName!="ConsoleLogin" | stats count by userIdentity.sessionContext.attributes.mfaAuthenticated

✓ 6,554 events (before 11/28/25 12:58:23.000 PM) No Event Sampling ▼

Events Patterns Statistics (1) Visualization

Show: 20 Per Page ▼ Format ▼ Preview: On

userIdentity.sessionContext.attributes.mfaAuthenticated	count
false	2155

4.3 Web Server Asset Identification (Question 3)

Objective: Characterize the hardware profile of the web servers.

Analysis: Using sourcetype="hardware", I identified hosts named gacrux. To verify that the 'gacrux' endpoints identified in the hardware logs were indeed the web servers, I analysed the naming convention and cross-referenced the hostname with the *access_combined* sourcetype. This revealed that these hosts were generating Apache web logs. Furthermore, I observed high-frequency HTTP GET requests from the User-Agent ELB-HealthChecker/2.0. This specific traffic

pattern confirms that these instances are registered targets behind an AWS Elastic Load Balancer [5], actively serving HTTP traffic.

Finding: I found the CPU_TYPE listed as Intel Xeon CPU E5-2676 v3. Mapping hardware profiles and hostnames to web-facing services gives the SOC an accurate asset inventory, which is essential for scoping incidents, correlating web logs to specific servers, and spotting performance anomalies that may indicate attacks such as cryptojacking or DoS.

Query:

```
index=botsv3 sourcetype="hardware"
```

```
index=botsv3 sourcetype="access_combined" host="gacrux.i-0920036c8ca91e501" http*
```

← → ↻ Not Secure http://student-vm:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dbotsv3

splunk>enterprise Apps ▾

Search Analytics Datasets Reports Alerts Dashboards

New Search

index=botsv3 sourcetype="hardware"

✓ 3 events (before 11/28/25 1:20:08.000 PM) No Event Sampling ▾

Events (3) Patterns Statistics Visualization

Timeline format ▾ — Zoom Out + Zoom to Selection × Deselect

Format ▾ Show: 20 Per Page ▾ View: List ▾

< Hide Fields	≡ All Fields	i	Time	Event
SELECTED FIELDS a host 3 a source 1 a sourcetype 1	INTERESTING FIELDS a index 1 # linecount 1 a punct 1 a splunk_server 1 a timestamp 1 + Extract New Fields	>	8/20/18 3:26:25.000 PM	KEY VALUE CPU_TYPE Intel(R) Xeon(R) CPU E5-2676 v3 @ 2.40GHz CPU_CACHE 30720 KB CPU_COUNT 2 HARD_DRIVES xvda 8 GB; Show all 9 lines host = gacrux.i-09cbc261e84259b54 source = hardware sourcetype = hardware
		>	8/20/18 3:24:24.000 PM	KEY VALUE CPU_TYPE Intel(R) Xeon(R) CPU E5-2676 v3 @ 2.40GHz CPU_CACHE 30720 KB CPU_COUNT 2 HARD_DRIVES xvda 8 GB; Show all 9 lines host = gacrux.i-06fea586f3d3c8ce8 source = hardware sourcetype = hardware
		>	8/20/18 2:34:50.000 PM	KEY VALUE CPU_TYPE Intel(R) Xeon(R) CPU E5-2676 v3 @ 2.40GHz CPU_CACHE 30720 KB CPU_COUNT 2 HARD_DRIVES xvda 8 GB; Show all 9 lines host = gacrux.i-0cc93bade2b3cba63 source = hardware sourcetype = hardware

New Search

Save As

index=botsv3 sourcetype="access_combined" host="gacrux.i-0920036c8ca91e501" http

✓ 2,772 events (before 11/28/25 7:47:48.000 PM) No Event Sampling ▾ Job ▾ II ▮

Events (2,772) Patterns Statistics Visualization

Timeline format ▾ — Zoom Out + Zoom to Selection × Deselect

Format ▾ Show: 20 Per Page ▾ View: List ▾ < Prev 1 2 3 4

< Hide Fields	≡ All Fields	i	Time	Event
SELECTED FIELDS a host 1 a source 1 a sourcetype 1		>	8/20/18 4:17:39.000 PM	172.16.0.149 - - [20/Aug/2018:15:17:39 +0000] "GET / HTTP/1.1" 200 19591 "-" "ELB-HealthChecker/2.0" host = gacrux.i-0920036c8ca91e501 source = /var/log/httpd/access_log sourcetype = access_combined
		>	8/20/18 4:17:39.000 PM	172.16.1.239 - - [20/Aug/2018:15:17:38 +0000] "GET / HTTP/1.1" 200 19591 "-" "ELB-HealthChecker/2.0"

4.4 S3 Bucket Permissions Analysis (Question 4)

Objective: Identify the specific API call that exposed data to the public.

Analysis: I began by searching for eventName='PutBucketAcl', which tracks changes to S3 bucket permissions. This returned two events. Instead of guessing, I analysed the 'Interesting Fields' sidebar to find parameters related to access control.

Finding: I discovered the field requestParameters.AccessControlPolicy.AccessControlList.Grant{}.Grantee.URI. Checking the values for this field, I spotted <http://acs.amazonaws.com/groups/global/AllUsers> - this AWS

identifier for public access. I added this to my search to confirm the single relevant event, pointing to Event ID ab45689d-69cd-41e7-8705-5350402cf7ac.

While detecting PutBucketAcl events allows for reactive containment, the fact that this command succeeded reveals a systemic governance failure. Modern AWS environments utilize "S3 Block Public Access" settings at the account level to override such ACLs. The success of this action proves that Frothly's SOC lacks a Cloud Security Posture Management (CSPM) tool [7] to enforce baseline compliance, allowing a simple configuration error to escalate into a public data breach

Query:

```
index=botsv3 sourcetype="aws:cloudtrail" eventName="PutBucketAcl"
```

```
requestParameters.AccessControlPolicy.AccessControlList.Grant{}.Grantee.URI="http://acs.amazonaws.com/groups/global/AllUsers"
```

← → ↻ Not Secure http://student-vm

splunk>enterprise Apps ▾

Search Analytics Datasets Reports Alerts Dashboards

New Search

index=botsv3 sourcetype="aws:cloudtrail" eventName="PutBucketAcl"

✓ 2 events (before 11/28/25 1:35:27.000 PM) No Event Sampling ▾

Events (2) Patterns Statistics Visualization

requestParameters.AccessControlPolicy.AccessControlList.Gra...

2 Values, 100% of events Selected Yes No

Reports

Top values Top values by time Rare values

Events with this field

Values	Count	%
http://acs.amazonaws.com/groups/s3/LogDelivery	6	300%
http://acs.amazonaws.com/groups/global/AllUsers	2	100%

← → ↻ Not Secure http://student-vm:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dbotsv3 sourcetype%3D%3Daws%3Acloudtrail% eventName%3D%3DPutBucketAcl

New Search

index=botsv3 sourcetype="aws:cloudtrail" eventName="PutBucketAcl" "requestParameters.AccessControlPolicy.AccessControlList.Grant().Grantee.URI"="http://acs.amazonaws.com/groups/global/AllUsers"

✓ 1 event (before 11/28/25 1:33:57.000 PM) No Event Sampling ▾

Events (1) Patterns Statistics Visualization

Timeline format ▾ — Zoom Out + Zoom to Selection X Deselect

Format ▾ Show: 20 Per Page ▾ View: List ▾

< Hide Fields All Fields

SELECTED FIELDS

- # host 1
- # source 1
- # sourcetype 1

INTERESTING FIELDS

- # awsRegion 1
- # date_hour 1
- # date_mday 1
- # date_minute 1
- # date_month 1
- # date_second 1
- # date_wday 1
- # date_year 1
- # date_zone 1
- # eventId 1
- # eventName 1
- # eventSource 1
- # eventTime 1
- # eventType 1
- # eventVersion 1

Time Event

8/20/18 2:01:46.000 PM

```
[{"awsRegion": "us-west-1", "eventId": "ab45689d-69cd-41e7-8705-5350402cf7ac", "eventName": "PutBucketAcl", "eventSource": "s3.amazonaws.com", "eventTime": "2018-08-20T13:01:46Z", "eventType": "AwsApiCall", "eventVersion": "1.05", "recipientAccountId": "622676721278", "requestId": "4874880003569438", "requestParameters": {}, "responseElements": null, "sourceIPAddress": "107.77.212.175", "userAgent": "signin.amazonaws.com", "userIdentity": {}}]
```

Show as raw text

host = splunk.frothyly source = s3://cloudtrail-622676721278/AWSLogs/622676721278/CloudTrail/us-west-1/201... sourcetype = aws:cloudtrail

4.5 Attribution of Actions (Question 5)

Objective: Determine the actor responsible for the configuration change.

Analysis: Examining the userIdentity object within the Event ID identified in Section 4.4.

Finding: The user bstoll (Bud) executed the command. This allows the SOC to pivot the investigation: is Bud a malicious insider, or were his non-MFA credentials (identified in Section 4.2) compromised?

```
8/20/18      { [-]
2:01:46.000 PM  awsRegion: us-west-1
                eventID: ab45689d-69cd-41e7-8705-5350402cf7ac
                eventName: PutBucketAcl
                eventSource: s3.amazonaws.com
                eventTime: 2018-08-20T13:01:46Z
                eventType: AwsApiCall
                eventVersion: 1.05
                recipientAccountId: 622676721278
                requestID: 4874880003569438
                requestParameters: { [+]}
                responseElements: null
                sourceIPAddress: 107.77.212.175
                userAgent: signin.amazonaws.com
                userIdentity: { [-]
                    accessKeyId: ASIAZB6TMXZ70A2RDK5X
                    accountId: 622676721278
                    arn: arn:aws:iam::622676721278:user/bstoll
                    invokedBy: signin.amazonaws.com
                    principalId: AIDAJUFKXZ44LV4EN4MGK
                    sessionContext: { [-]
                        attributes: { [-]
                            creationDate: 2018-08-20T12:19:44Z
                            mfaAuthenticated: false
                        }
                    }
                    type: IAMUser
                    userName: bstoll
                }
            }
        Show as raw text
        host = splunk.froth.ly | source = s3://cloudtrail-622676721278/AWSLoos/622676721278/CloudTrail/us-west-1/201... | sourcetvpe = aws:cloudtrail
```

4.6 Data Exposure Scope (Question 6)

Objective: Identify the exposed resource.

Analysis: The requestParameters field in the log entry specified the target bucket.

Finding: The bucket name is frothlywebcode which implies source code exposure, risking API keys or IP theft (MITRE ATT&CK T1530 [3]). The identification allows the SOC to estimate business impact and prioritise remediation.

```

8/20/18      { [-]
2:01:46.000 PM  awsRegion: us-west-1
                eventID: ab45689d-69cd-41e7-8705-5350402cf7ac
                eventName: PutBucketAcl
                eventSource: s3.amazonaws.com
                eventTime: 2018-08-20T13:01:46Z
                eventType: AwsApiCall
                eventVersion: 1.05
                recipientAccountId: 622676721278
                requestID: 487488D003569438
                requestParameters: { [-]
                                    AccessControlPolicy: { [-]
                                        AccessControlList: { [-]
                                            Grant: [ [-]
                                                { [+]
                                                }
                                                { [+]
                                                }
                                                { [+]
                                                }
                                                { [+]
                                                }
                                                { [+]
                                                }
                                                { [+]
                                                }
                                                { [+]
                                                }
                                            ]
                                        }
                                    }
                Owner: { [+]
                }
                xmlns: http://s3.amazonaws.com/doc/2006-03-01/
            }
            acl: [ [+]
            ]
            bucketName: frothlywebcode

```

4.7 Unauthorized Artifact Upload (Question 7)

Objective: Confirm if the exposure was exploited.

Analysis: I queried aws:s3:accesslogs for the .txt extension and PUT method to find successful uploads by external parties.

Finding: A file named OPEN_BUCKET_PLEASE_FIX.txt was uploaded. The filename suggests a "Gray Hat" security researcher. Confirming that external actors were able to write into the bucket proves that integrity was breached, which helps the SOC justify immediate containment, deeper threat hunting for potential payloads, and long-term hardening of cloud storage controls.

Query:

index=botsv3 sourcetype="aws:s3:accesslogs" .txt PUT

New Search

index=botsv3 sourcetype="aws:s3:accesslogs" * .txt *PUT*

Time range: All time

1 event (before 11/28/25 2:08:28.000 PM) No Event Sampling

Events (1) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

1 millisecond per column

Format Show: 20 Per Page View: List

	Time	Event
>	8/20/18 2:02:44.000 PM	4c018853e740f450eb45f68c9f5eff634745488ae540130432c9f64fae310d frothywebcode [20/Aug/2018:13:02:44 +0000] 52.66.146.128 - DF18A980E236904 REST.PUT.OBJECT OPEN_BUCKET_PLEASE_FIX.txt "PUT /OPEN_BK CKET_PLEASE_FIX.txt HTTP/1.1" 200 - - 377 268 9 "-" "Boto3/1.7.62 Python/2.7.14 Linux/4.14.47-64.38.amzn2.x86_64 Botocore/1.8.12" - host = splunk.frothy.ly source = s3://frothyweblogs/s32018-07-26-01-20-56-19D73C05AA29AEDB sourcetype = aws:s3:accesslogs

4.8 Endpoint Anomalies (Question 8)

Objective: Detect configuration drift in endpoints.

Analysis: Using sourcetype="WinHostMon", I grouped hosts by OS version. However, the standard host field only provided the short hostname, leaving the domain uncertain. To resolve this, I performed some investigative digging through the raw log entries for BSTOLL-L.

Finding: The ComputerName field reveals the full path BSTOLL-L.frothy.ly; unlike most hosts running Windows 10 Pro, this endpoint runs Windows 10 Enterprise, indicating an administrator workstation linked to the same user and timeframe as the risky cloud activity. This makes it a high-value asset and priority for SOC containment. If compromised, BSTOLL-L.frothy.ly could hold credentials enabling lateral movement [4] across the domain, escalating the breach from a cloud misconfiguration to a full domain compromise.

Query:

index=botsv3 sourcetype="WinHostMon" | stats count by host, OS Version

index=botsv3 host="BSTOLL-L" computername

New Search

index=botsv3 sourcetype="winhostmon" | stats count by host, OS Version

✓ 129,679 events (before 11/28/25 2:20:59.000 PM) No Event Sampling ▼

Events Patterns **Statistics (8)** Visualization

Show: 20 Per Page Format Preview: On

host ↕	OS ↕	Version ↕
ABUNGST-L	Microsoft Windows 10 Pro	10.0.17134
BGIST-L	Microsoft Windows 10 Pro	10.0.17134
BSTOLL-L	Microsoft Windows 10 Enterprise	10.0.17134
BTUN-L	Microsoft Windows 10 Pro	10.0.17134
FYODOR-L	Microsoft Windows 10 Pro	10.0.17134
JWORTOS-L	Microsoft Windows 10 Pro	10.0.17134
MKRAEUS-L	Microsoft Windows 10 Pro	10.0.17134
PCERF-L	Microsoft Windows 10 Pro	10.0.17134

Nov 29 16:52

Search | Splunk 10.0.2 +

splunk>enterprise Apps Administrator

Search Analytics Datasets Reports Alerts Dashboards

New Search

index=botsv3 host="BSTOLL-L" | computername

✓ 24,457 events (before 11/29/25 4:52:03.000 PM) No Event Sampling ▼

Events (24,457) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

Format Show: 20 Per Page View: List

< Hide Fields All Fields	i Time	Event
SELECTED FIELDS a host 1 a source 6 a sourcetype 6	> 8/20/18 4:17:58.000 PM	08/20/2018 03:17:58 AM ... 3 lines omitted ... EventType=0 Type=Information ComputerName=BSTOLL-L.froth.ly TaskCategory=Process Termination Show all 21 lines
INTERESTING FIELDS a Account_Domain 6	host = BSTOLL-L source = WinEventLog:Security sourcetype = WinEventLog:Security	

enforce MFA on all privileged API usage, and treat administrator endpoints such as BSTOLL-L.froth.ly as high-value assets needing additional monitoring and control.

5.1 Immediate containment (0–24 hours)

- Block public access on frothlywebcode and verify no other buckets grant *AllUsers/AuthenticatedUsers* ACLs.
- Rotate credentials for bstoll and related IAM roles, and remove unapproved uploads such as *OPEN_BUCKET_PLEASE_FIX.txt* after preserving evidence.

5.2 Short-term recovery (next 1–2 weeks)

- Enforce an IAM condition that denies API actions unless MFA is true for all human users and high-risk roles.
- Isolate BSTOLL-L.froth.ly for full forensic analysis, including malware scanning, credential dump checks, and review of administrative tool usage.
- Develop and deploy rules that alert on *PutBucketAcl* granting public access, high-volume non-MFA API activity, and changes originating from privileged endpoints.

5.3 Long-term SOC improvements (1–3 months)

- Implement a CSPM capability to continuously detect and auto-remediate public S3 buckets and other misconfigurations [7].
- Conduct a least-privilege review of IAM users and roles, ensuring developers cannot modify global ACLs without change control and dual authorisation.
- Integrate scenarios like this BOTSv3 incident into SOC runbooks and training so Tier 1–3 analysts can rapidly recognise similar patterns and execute coordinated response and recovery.

6. References

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- [3] Borges, D. and @1njection (2017) Remote Services: SMB/windows admin shares, Remote Services: SMB/Windows Admin Shares, Sub-technique T1021.002 - Enterprise | MITRE ATT&CK®. Available at: <https://attack.cloudfall.cn/techniques/T1021/002/> (Accessed: 07 December 2025).
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- [5] (No date) How elb works - elastic load balancing. Available at: <https://docs.aws.amazon.com/elasticloadbalancing/latest/userguide/how-elastic-load-balancing-works.html> (Accessed: 07 December 2025).
- [6] G, Alistair. (22/01/2025) How attackers bypass multi-factor authentication. Available at: <https://www.bugcrowd.com/blog/mfa-security-part-1-how-attackers-bypass-multi-factor-authentication/> (Accessed: 07 December 2025).
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7. Appendix A: Video Presentation

<https://youtu.be/WJN0MkBlE7c>

8. Appendix B: Generative AI Declaration

Student Declaration of AI Tool use in this Assessment

Solo Work	S1 - Generative AI tools have not been used for this assessment.	☐
Assisted Work	A1 – Idea Generation and Problem Exploration Used to generate project ideas, explore different approaches to solving a problem, or suggest features for software or systems. Students must critically assess AI-generated suggestions and ensure their own intellectual contributions are central.	☐
	A2 - Planning & Structuring Projects AI may help outline the structure of reports, documentation and projects. The final structure and implementation must be the student's own work.	☐
	A3 – Code Architecture AI tools maybe used to help outline code architecture (e.g. suggesting class hierarchies or module breakdowns). The final code structure must be the student's own work.	☐
	A4 – Research Assistance Used to locate and summarise relevant articles, academic papers, technical documentation, or online resources (e.g. Stack Overflow, GitHub discussions). The interpretation and integration of research into the assignment remain the student's responsibility.	☐
	A5 - Language Refinement Used to check grammar, refine language, improve sentence structure in documentation not code. AI should be used only to provide suggestions for improvement. Students must ensure that the documentation accurately reflects the code and is technically correct.	☐
	A6 – Code Review AI tools can be used to check comments within the code and to suggest improvements to code readability, structure or syntax. AI should be used only to provide suggestions for improvement. Students must ensure that the code accurately reflects their knowledge and is technically correct.	☐

	A7 - Code Generation for Learning Purposes Used to generate example code snippets to understand syntax, explore alternative implementations, or learn new programming paradigms. Students must not submit AI-generated code as their own and must be able to explain how it works.	☐
	A8 - Technical Guidance & Debugging Support AI tools can be used to explain algorithms, programming concepts, or debugging strategies. Students may also help interpret error messages or suggest possible fixes. However, students must write, test, and debug their own code independently and understand all solutions submitted.	☐
	A9 - Testing and Validation Support AI may assist in generating test cases, validating outputs, or suggesting edge cases for software testing. Students are responsible for designing comprehensive test plans and interpreting test results.	☐
	A10 - Data Analysis and Visualization Guidance AI tools can help suggest ways to analyse datasets or visualize results (e.g. recommending chart types or statistical methods). Students must perform the analysis themselves and understand the implications of the results.	☐
	A11 - Other uses not listed above Please specify:	☐
Partnered Work	P1 - Generative AI tool usage has been used integrally for this assessment Students can adopt approaches that are compliant with instructions in the assessment brief. Please Specify: • Research further concepts • Write/generate/troubleshoot scripts to facilitate analysis • Report generation and improvements • Readme report, script crafting.	☒

Please provide details of AI usage and which elements of the coursework this relates to:
SPL queries, rewording readme report, research of splunk understanding.

I understand that the ownership and responsibility for the academic integrity of this submitted assessment falls with me, the student.	☒
I confirm that all details provide above are an accurate description of how AI was used for this assessment.	☒